LIS-SRS-001 · VERSION 1.1 · DRAFT FOR REVIEW · 2026-10-14

Sky LIS — Business Software Requirements Specification

Prepared for	National Technology — LIS project	**Prepared by**	Claude Code (AI analyst), for review by Mohamed Hashem
Classification	Internal — Confidential	**Basis**	Client Portal prototype v2 (full scope) and v2.1 (release-1 scope) · Super Admin Portal prototype v3 (full scope) and v3.1 (release-1 scope) · LIS Requirements Specification LIS-RS-001 · Prototype comparison LIS-PC-001 and build plan LIS-BP-001 · LDM 9.2/9.3 knowledge base (domain behaviour)

2 Portals

16 Modules

68 Client pages (30 in R1)

21 Admin pages (17 in R1)

12 Workflows

27 Business rules

Version	Date	Author	Change
1.0	2026-10-09	Claude Code	First complete business SRS covering both portals, all modules, pages and workflows, with the release-1 scope matrix.
1.1	2026-10-14	Claude Code	CP-41 Tests: parent services on the test page (the services carrying the test; join or leave profiles and cultures from the test, the other types set up from the service), as the Super Admin prototype shows. Page ids unchanged.

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1. Introduction

1.1 Purpose

This document specifies, in business terms, what the Sky LIS product does: the two portals, their modules and pages, the data each page handles, the actions users can take, the rules the system enforces, and the end-to-end workflows that connect the pages. It is the reference for product owners, laboratory domain experts, designers, developers and testers, and the baseline for acceptance.

1.2 Scope

Sky LIS is a cloud, subscription-based, multi-tenant Laboratory Information System operated by National Technology. The Super Admin Portal serves National Technology staff; the Client Portal serves each subscribed laboratory (tenant) with its branches and staff. This SRS describes the full target product and marks the subset delivered in Release 1.

Out of scope:

- Installation, hosting and infrastructure procedures.
- Detailed visual design (the prototypes are the UI reference).
- Quality control, outsourcing to external laboratories, SMS campaigns and inter-branch transfer modules beyond the placeholders noted here.
- Country-specific regulatory reporting.

1.3 Conventions

Convention	Meaning
Page ID	CP-nn for Client Portal pages, SA-nn for Super Admin pages. IDs are stable.
Release	R1 = included in Release 1 (prototypes v2.1 / v3.1). R2 = later release (full-scope prototypes v2 / v3).
Business rule	BR-nn, listed in section 9 and referenced from pages and workflows.
Workflow	WF-nn, step tables with actor, action and system response.
Requirement reference	Codes such as REG-005 refer to the functional requirements in LIS-RS-001.

2. Business overview

2.1 Business context

National Technology replaces its on-premise LDM laboratory system with Sky LIS, a cloud product sold by subscription. A laboratory subscribes online, receives its own portal address and an independent copy of the global laboratory master data, and runs its daily operation — reception, samples, results, printing, money — from the Client Portal across its branches. National Technology operates the platform from the Super Admin Portal: client lifecycle, plans and wallets, global master data, libraries and platform configuration.

2.2 Business goals

1. Let a laboratory start operating within a day of subscribing, with a ready catalogue and price lists.
2. Preserve every operational safeguard laboratories rely on: duplicate-patient protection, the sample Received gate, verified-only printing, delta and panic alerts, contract-correct pricing, auditable money movements.
3. Serve reception, bench, sign-off, accounts and administration personas from one bilingual, responsive workspace.
4. Connect analysers and PCR instruments and release normal results automatically where the laboratory chooses to.
5. Give National Technology central control of master data, subscriptions and support.

2.3 Actors

Actor	Responsibilities
Super Admin / Operations Admin	National Technology staff: clients, plans and wallets, admin users and roles, global master data, libraries, platform parameters, sessions and audit.
Lab Administrator	Owns the tenant configuration: branches, roles, users, catalogue copy, payers and prices, treasuries, parameters, analysers and automation.
Reception clerk / cashier	Registers patients and visits, orders services, answers preparation questions, collects money, prints labels, job orders, receipts and verified reports, closes the drawer.
Phlebotomist / courier	Collects samples in the laboratory or at home, is recorded as collector, brings lab-to-lab samples.
Technician / chemist	Receives samples, runs the worklist, enters or imports results, reviews, rejects unfit samples.
Unit head / responsible doctor	Verifies (e-signs) results, handles abnormal, panic and delta-flagged results, releases reports, owns auto-verification rules.
Microbiologist / pathologist	Enters culture results and formatted-text reports.
Accounts officer	Price lists and contracts, refunds and transfers, claims, financial reports.
Integration engineer	Analyser definitions and mappings, site agent, PCR kits.
Patient / payer / referring doctor	External parties receiving quotes, receipts, reports and claims.

2.4 Key identifiers

Identifier	Meaning
Patient No.	Permanent file number, auto-generated, never changes (P0004912).
Accession	Visit number: date-based sequence per tenant (2610080001).
Sample ID	Unique barcode number per tube, never reused (S2610080001).
Booking No. / Batch No.	Home-visit booking and batch (batch / lab-to-lab) grouping numbers.
Print list No. / Claim No. / Receipt No.	Numbers printed on reports, claims and receipts.

2.5 Release scope summary

Release 1 delivers the cross-cutting platform plus every page that existed in the first prototypes with its enhancements (Client Portal v2.1) and the complete operator portal except the integration library (Super Admin v3.1). Release 2 adds the new client pages and the integration, automation and quality-control modules. In Release 1 registration omits the hospital Location/Department and Room fields and captures the payment before saving; samples are received automatically on save because the sample pages arrive in Release 2.

3. System overview

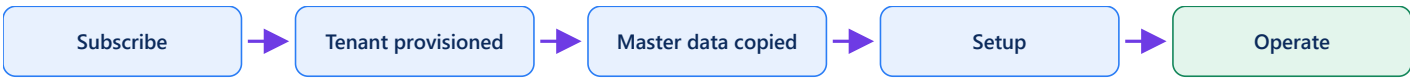
3.1 Two portals, one platform

Super Admin Portal

National Technology operates the platform: clients, plans and wallets, admin users and roles, global master data, integration libraries, platform parameters, sessions and audit.

Client Portal

Each laboratory runs reception, samples, laboratory, finance, setup, integration, automation and account from its own portal URL, across its branches, in Arabic or English.



Global master data flows from the Super Admin Portal into each tenant at creation (BR-25).

3.2 Cross-cutting capabilities

Capability	Description
Tenancy & branches	Every record belongs to one tenant; users work in one branch per session; all transactions are branch-stamped. Global master data is copied to the tenant at creation and maintained independently afterwards (BR-25).
Bilingual, RTL, responsive	Arabic and English on every screen with true right-to-left layout; master data carries Arabic names; documents print in the configured language; desktop, tablet and phone layouts.
Security	Username/password, OTP for recovery and credential changes, password policy, session timeout, lock and expiry, roles with per-screen Query/Insert/Update/Delete, special rights, result rights per clinical group and branch, device-bound treasuries.
Audit & notifications	Every create, update, delete, status change, print, payment, refund, transfer, cancellation and privilege change is logged with user, time, branch and device; a notification centre surfaces panic values, delays, treasury requests and agent status.
Grids & editors	Search, status and type filters, Excel export, pagination, row actions; editors validate mandatory fields, block deletion of referenced records and prefer deactivation.
Documents	Barcode labels (Code 128), job order, receipt, invoice, refund receipt, price quote, laboratory report with template, verifier signature and QR, abbreviated worksheet, worklist, cash-closing sheet, claim, analyser code sheet.
Keyboard	Ctrl+S saves the open form, Ctrl+K focuses search, F2 opens the service picker, Enter adds a typed service or shortcut, Esc closes dialogs.

3.3 Modules

Client Portal

Module	Description	Release
Access & onboarding	Public website, self-service subscription, login, password recovery, branch selection.	R1
Overview	Operational dashboard for the branch.	R1
Reception	Patient files and visits in four registration modes, search, quotes, printing of verified reports.	R1 (walk-in, search, patients, quote) · R2 (home, batch, lab-to-lab, print reports)
Samples	Tube lifecycle: collection, receipt, rejection, replacement, tracking.	R2
Laboratory	Worklist, result entry and sign-off, status work queue, quality control placeholder.	R1 (worklist, results by accession) · R2 (results by sample, work queue, QC)
Finance	Payers and pricing, payments and money movements, treasuries, closing, claims and reports.	R1 (payments, treasuries, payers, base price lists) · R2 (rest)
Setup	Tenant configuration and catalogue.	R1 (20 pages) · R2 (catalogue depth, parameters)
Integration	Analysers, code mapping, site agent, PCR kits.	R2
Automation	Auto-verification settings and medical rules.	R2
Account	Profile, wallet, support, audit log.	R1 (profile) · R2 (wallet, support, audit)

Super Admin Portal

Module	Description	Release
Overview	Home and platform dashboard.	R1
Administration	Clients, subscriptions and wallets, roles, admin users, sessions, audit log.	R1
Master data	Global laboratory catalogue and lists copied to tenants.	R1
Microbiology	Drug types, commercial names, antibiotics, organisms, sensitivity values.	R1
Integration library	Analyser models, PCR kits, code systems, rule templates.	R2
Platform	Platform parameters, notification templates, administrator profile.	R1

4. Client Portal — pages

> Each page card states its purpose, who uses it, the data it handles, the actions available, the business rules it enforces, the pages or master data it depends on, and its release.

4.1 Access & onboarding (4)

CP-01 **Public website** Access & onboarding R1

Presents the product and routes visitors to login or subscription.

Actors	Prospective laboratory
Key data	Product features, demo KPIs.
Actions	Login; Subscribe now; language switch.
Business rules	None.
Depends on	—

CP-02 **Subscription wizard** Access & onboarding R1

Creates a new tenant in four steps: lab details, OTP to the manager mobile, administrator credentials, portal URL delivery.

Actors	Laboratory manager
Key data	Lab name (EN/AR), country, governorate, address, mobile, medical licence, manager name and mobile, email; username, password.
Actions	Send OTP; Verify; Create account; Login.
Business rules	Mobile validated by country length; password policy (8+, upper, lower, digit, special, match); OTP valid 5 minutes; tenant receives an independent copy of global master data (BR-25).
Depends on	Countries & governorates (SA) Notification templates (SA)

CP-03 **Login & password recovery** Access & onboarding R1

Authenticates users and recovers credentials with OTP.

Actors	All users
Key data	Username, password; username or mobile for recovery.
Actions	Login; Forgot password → OTP → new credentials; show/hide password; remember me.
Business rules	Lockout after repeated failures; locked or expired users refused; credentials updated only after OTP.
Depends on	Users, Notification gateway

CP-04 **Branch selection** Access & onboarding R1

Binds the session to one of the user's branches.

Actors	All users
Key data	Assigned branches.
Actions	Continue.
Business rules	Only assigned branches listed; every transaction of the session is stamped with the branch.
Depends on	Branches, Users

4.2 Overview (1)

CP-05 **Dashboard** Overview R1

Shows the branch's day at a glance and offers quick actions.

Actors	All users
Key data	KPIs: registrations, collected today, samples received*, pending results, unverified backlog, delayed tests, rejected samples*, wallet; 7-day registrations chart; delayed-tests table; recent activity. (*R2)
Actions	Quick actions to the main pages; refresh.
Business rules	Figures are computed for the session branch and day; delays derive from test durations (BR-13).
Depends on	Results, Samples, Audit log

4.3 Reception (8)

CP-06 **Registration** Reception R1 without Location/Department and Room; payment captured before saving

Registers a walk-in patient and visit: patient file, visit data, ordered services, preparation answers, discount, samples and documents; in Release 1 also the payment.

Actors	Reception clerk
Key data	Patient: auto number, title, name, gender, date of birth / age, condition, mobile, national ID, nationality, email, governorate, city, address, comments, VIP. Visit: branch, location type, (R2: location/department, room), treatment no., referring doctor, payer, contract tier, insurance ID, diagnosis, urgent, biohazard, collected outside. Lines: service, status Ordered/Reserved, expected result, patient share, payer share, discount, collector. Preparation answers. Discount amount/% and reason. R1: payment method, amount, reference.
Actions	Search existing patient; new patient; add services by name, alias, short code, shortcut number, shortcut set or picker; set Reserved; remove line; apply discount; answer preparations; R1: choose payment method and amount; Save (Ctrl+S); edit a saved visit.
Business rules	Similar-name warning compares name and mobile (BR-01, REG-002); title sets gender; age ↔ DOB; gender-restricted services refused; frequency check alerts or blocks repeats (BR-12); contract alert conditions shown; auto-add rules add companion services; discount within the user cap with mandatory reason; tubes generated per group + sample type + condition (BR-02); Reserved keeps the line in the bill without a sample (BR-05); payer and tier set every price (BR-06); R1: paid amount ≤ amount due, reference mandatory for non-cash; save prints labels and job order and the receipt per parameters.
Depends on	Services & tests, Payers & contracts, Contract pricing, Patient conditions, Preparations, Diagnoses, Shortcut sets, Test rules, Doctors, Reasons, Parameters, Treasuries

CP-07 **Home reservation** Reception R2

Books a home visit and converts it into a registration when the sample returns.

Actors	Reception clerk, phlebotomist
Key data	Booking no., patient data, mandatory address, visit time, assigned phlebotomist, payer, services, quote, notes, status Booked/Converted/Cancelled.
Actions	New booking; preliminary labels and job order; cancel; convert to visit.
Business rules	Address mandatory; conversion creates the visit without retyping and marks the booking Converted.
Depends on	Registration, Users (phlebotomists), Services

CP-08 **Batch registration** Reception R2

Registers a group of people on one contract with the same services.

Actors	Reception clerk
Key data	Company/group, payer and tier, doctor; member rows (patient no., title, name, gender, age, condition); service set; batch no.
Actions	Add row; look up patient number; choose services once; Save batch; go to batch payment.
Business rules	One accession per member under one batch number; duplicate check per row; documents printed for all members.
Depends on	Registration, Payers, Batch payment

CP-09 **Lab-to-Lab registration** Reception R2

Registers samples arriving from a contracted laboratory.

Actors	Reception clerk
Key data	Sending lab (fills payer, tier, courier, phone), rows with patient data and services, collected-inside flag, batch no.
Actions	Choose lab; add rows; pick services per row; Save batch.
Business rules	Unticked "collected inside" marks the tube drawn outside and prints it on the job order; billing goes to the sending lab through claims.
Depends on	Sending labs, Registration, Claims

CP-10 **Search registrations** Reception R1

Finds visits and acts on them.

Actors	Reception clerk, cashier
Key data	Accession, patient no., name, mobile, date range, branch, payer, status; row: services with reserved/cancelled styling, patient share, remaining balance.
Actions	Edit visit; samples (dialog in R1, page in R2); payments; documents (receipt, job order, labels, report); results; cancel services with reason.
Business rules	Verified services cannot be cancelled; cancellation keeps the record with reason and user (BR-15).
Depends on	Registration, Payments, Documents, Reasons

CP-11 **Patients** Reception R1

Patient files, history and documents.

Actors	Reception clerk, lab administrator
Key data	Patient no., name, gender, age, mobile, condition, VIP, visit count, last visit; documents.
Actions	Search by number, name, mobile or national ID; visit history with results; attach documents; merge duplicates (right); new visit.
Business rules	Merge moves visits, results and payments to the surviving file and is audited; documents visible from every branch.
Depends on	Registration, Results, Audit log

CP-12 **Price query** Reception R1

Answers "how much, when, and must I fast?" without registering.

Actors	Reception clerk
Key data	Payer, tier, services; list price, patient share, payer share, expected result, preparation.
Actions	Add services; print quote; continue to registration.
Business rules	Nothing is saved until the user continues to registration; prices come from contract pricing (BR-07).
Depends on	Contract pricing, Services

CP-13 **Print reports** Reception R2

Prints the day's verified reports in bulk.

Actors	Reception clerk
Key data	Verified-unprinted report units, print list, print list no.
Actions	Add rows or whole accessions to the list; print list; abbreviated worksheet.
Business rules	Only verified reports print (BR-09); each print list gets a number printed on the reports.
Depends on	Work queue, Results

4.4 Samples (4)

CP-14 **Collection & receipt** Samples R2 in Release 1 samples are received automatically on save

Moves tubes through Ordered → Collected → Received by barcode.

Actors	Phlebotomist, technician
Key data	Sample ID, accession, patient, unit, type, condition, services, status, events.
Actions	Scan in collect or receive mode; collect/receive all samples of an accession; today's list.
Business rules	Received gate: results cannot be entered before receipt (BR-03); auto-collect/receive parameters emulate the gate for non-scanning laboratories.
Depends on	Registration (samples) Parameters

CP-15 **Samples by visit** Samples R2

Lists a visit's tubes and reprints documents.

Actors	Reception, technician
Key data	Per tube: unit, type, condition, status, status date, ID, labels, services.
Actions	Print one label; print all labels; job order; track.
Business rules	—
Depends on	Registration

CP-16 **Reject / unreject** Samples R2

Rejects unfit tubes and reinstates or replaces them.

Actors	Technician
Key data	Sample ID, reason, accession, patient; rejected list.
Actions	Look up; reject with reason; unreject (same barcode); issue replacement (new barcode).
Business rules	Rejection needs a reason and notifies reception; replacement follows the tenant barcode policy (BR-18); rejected tubes accept no results.
Depends on	Reasons Labels Parameters

CP-17 **Sample tracking** Samples R2

Shows where a tube was and when.

Actors	Everyone
Key data	Timeline of status, location, user, time; time between stages; test round times.
Actions	Scan or search; open from worklist and results.
Business rules	—
Depends on	Samples, Results

4.5 Laboratory (5)

CP-18 **Worklist** Laboratory R1

Lists the tests to perform per unit, with turnaround status.

Actors	Technician, unit head
Key data	One row per test: patient no., accession, sample ID, patient, age/sex, test, test status, sample status, collection, expected result, delay, consumed %; filters by collection date, group, test, analyser, sample status, payer; status boxes; counters requests/tests/samples.
Actions	Search; print worklist; track sample; open results.
Business rules	Filtered by collection date; expected date = collection + duration counting test days; row colour by share of duration consumed (BR-13).
Depends on	Tests (timing), Samples, Parameters (thresholds)

CP-19 **Results by accession** Laboratory R1 filters auto-collapse, report units as tabs, no analyser column

Enters, reviews, verifies and prints results of a visit.

Actors	Technician, unit head, microbiologist, pathologist
Key data	Query: accession, patient no., name (%), test status, sample status, from analyser, not printed / not verified / delta. Case: patient data, CR, preparations, diagnosis, flags. One tab per report unit (clinical group) showing its sample IDs and status. Grid: test, result, unit, reference range, flag, sample status, test status, delta, printed, attachments, logs. Report comment, internal notes.
Actions	Run query (filters collapse once a visit is open); Save; Review; Verify; Undo; Preview; Print; insert comment from library; culture entry; text entry with templates; previous results; attachments; result log; status log.
Business rules	Received gate (BR-03); ranges resolved by sex, age, condition (BR-08); panic values alert; restricted lists accept listed values only; formulas compute automatically; delta flag must be acknowledged before verify (BR-11); verify needs a value and the result right; preview refused while values are missing; only verified prints (BR-09); printed flag and status log on print; edits of verified results need the special right.
Depends on	Tests, Roles (result rights), Comment library, Report templates, Parameters

CP-20 **Results by sample** Laboratory R2

Works only the tests on a scanned tube.

Actors	Technician
Key data	Sample ID → its report units (tabs).
Actions	Scan; same actions as results by accession.
Business rules	Same rules as CP-19.
Depends on	Results by accession, Samples

CP-21 **Work queue** Laboratory R2

Status-driven list of report units for the day's work and printing.

Actors	Technician, unit head, reception
Key data	Registration-date range, group, payer, from/to analyser, normal/abnormal, received-only, keep-previous; status boxes exact (Complete, Reviewed, Verified, Printed) and cumulative Not-X; nine colour states; count; print list.
Actions	Search; open report (R); review all; verify all; add to print list; print list; abbreviated report; analyser closing presets.
Business rules	Not Verified = Ordered + Complete + Reviewed, Not Reviewed = Ordered + Complete, Not Complete = Ordered (BR-10); bulk actions apply to every listed result and are audited; print list numbers.
Depends on	Results, Analysers, Roles

CP-22 **Quality control** Laboratory R2

Placeholder for QC materials, targets, Levey-Jennings and Westgard rules.

Actors	Technician, unit head
Key data	Analyser, test, level, value, target, SD index, rule, status.
Actions	View runs.
Business rules	Scope to be confirmed.
Depends on	Analysers, Tests

4.6 Finance (13)

CP-23 **Payments** Finance R1

Collects money on visits and prints receipts.

Actors	Cashier
Key data	Visit search; total, discount, patient share, paid, refunded, remaining; payment: amount, method (cash, card, POS, transfer), reference, treasury, device; receipts.
Actions	Pay (full or partial); refund when the balance is negative; receipts and invoice; reprint.
Business rules	Amount within the remaining balance; reference mandatory for non-cash; every payment stamped with treasury, cashier and device; receipt auto-printed per parameter.
Depends on	Treasuries, Treasury requests, Parameters

CP-24 Refunds

Finance

R2 refund action also available on Payments in R1

Returns money when a visit balance is negative.

Actors	Cashier with the Refund right
Key data	Negative balances; refund amount, method, reason; refund receipt.
Actions	Look up visit; refund; print receipt for signature.
Business rules	Refund never exceeds the refundable amount (BR-14); reason mandatory; audited.
Depends on	Payments, Reasons

CP-25 Money transfer

Finance

R2

Moves a visit to another contract or shifts a line between patient and payer.

Actors	Accounts, cashier with right
Key data	Visit, current and new payer/tier, re-priced lines, resulting balance; per-line actual amounts.
Actions	Contract transfer with preview; line transfer; apply.
Business rules	Re-pricing follows the new contract (BR-06); an overpayment becomes a refund; every transfer is logged.
Depends on	Payers, Contract pricing, Refunds

CP-26 Cancellations

Finance

R2 available as an action on Search registrations in R1

Cancels services of a visit with a reason.

Actors	Reception, accounts with right
Key data	Visit lines with status and shares; cancellation reason; cancelled list.
Actions	Select services; cancel.
Business rules	Verified services cannot be cancelled; related samples cancelled; receivables adjusted; nothing physically deleted (BR-15).
Depends on	Reasons, Samples

CP-27 **Batch payment** Finance R2

Settles a batch or lab-to-lab batch.

Actors	Cashier
Key data	Batch no., members with shares, paid and remaining.
Actions	Pay member; pay all.
Business rules	Lab-to-lab batches are billed to the sending lab through claims.
Depends on	Batch registration, Lab-to-Lab registration, Payments

CP-28 **Treasuries** Finance R1

Cash drawers per branch.

Actors	Lab administrator
Key data	Code, name (EN/AR), branch, status, approved users.
Actions	Add, edit, deactivate.
Business rules	Deactivation instead of deletion when referenced.
Depends on	Branches

CP-29 **Treasury requests** Finance R1

Grants treasury access to a user on a specific device.

Actors	Cashier (requests), manager (approves)
Key data	Requested by, branch, treasury, device ID, date, status Pending/Approved/Rejected.
Actions	Request from this device; approve; reject; revoke.
Business rules	A payment can only be taken on an approved treasury/device pair; decisions are audited.
Depends on	Treasuries, Users

CP-30 **Cash closing** Finance R2

Reconciles the drawer for a treasury, cashier and period.

Actors	Cashier, manager
Key data	Payments and refunds list; totals: payments, cash, card/POS/transfer, refunds, expected cash, outstanding balances, discounts.
Actions	Filter; print closing sheet.
Business rules	Expected cash = cash payments − refunds; card and transfer amounts are not in the drawer (BR-16).
Depends on	Payments, Refunds, Treasuries

CP-31 **Claims** Finance R2

Bills credit payers for a period.

Actors	Accounts
Key data	Payer, tier (or all), period, layout (standard, per tier, with results, totals only, with counts, credit note); claim no., visits, total, status Draft/Issued.
Actions	Preview; save draft; issue; print; Excel.
Business rules	Cash payers never receive claims; claim totals are the payer shares of the visits in the period.
Depends on	Payers, Contract pricing, Visits

CP-32 **Payers & contracts** Finance R1

Payers (ranks) and their contract tiers.

Actors	Accounts
Key data	Name (EN/AR), search key, type (cash, credit, insurance, lab), web-results portal credentials, tiers (EN/AR).
Actions	Add, edit; open contract pricing.
Business rules	Web username mandatory when the portal is enabled; payer with visits cannot be deleted.
Depends on	Contract pricing

CP-33 **Base price lists** Finance R1

Named lists holding the price per service.

Actors	Accounts
Key data	Code, name, year, active flag; price and active flag per service; derived contracts.
Actions	New list; edit prices inline; copy list with × factor or + amount; Apply.
Business rules	Contract prices refresh only on Apply (BR-07); inactive services do not appear at ordering.
Depends on	Services, Contract pricing

CP-34 **Contract pricing** Finance R2 Release 1 uses the seeded contract formulas

Derives each tier's prices from base lists by formula.

Actors	Accounts
Key data	Payer, tier, effective dates, up to three base lists with patient and payer factors, apply date; derived price grid with exclusion and alert condition per service.
Actions	Edit formula; Apply; exclude a negotiated price; set alert; copy from another tier; review report.
Business rules	Patient share + payer share per service (BR-04); excluded prices survive Apply; alerts pop up at ordering.
Depends on	Base price lists, Payers

CP-35 **Financial reports** Finance R2

Sixteen operational finance reports.

Actors	Accounts, management
Key data	Patients' income, payment methods, contract type, monthly statistics, patient statement, doctor income with share, clients income, service income analysis, outstanding balances, discounts, cancellations, transfers, total per day, price-list hand-out, cost vs income, turnaround compliance.
Actions	Run with period and branch; print; Excel.
Business rules	Figures reconcile with payments, refunds and visit shares.
Depends on	Visits, Payments, Doctors, Tests

4.7 Setup (22)

CP-36 **Branches** Setup R1

Laboratory branches.

Actors	Lab administrator
Key data	Code, EN/AR names, country, governorate, phone, email, currency, status.
Actions	Add, edit, deactivate.
Business rules	Branch with visits cannot be deleted.
Depends on	Countries & governorates

CP-37 **Roles & privileges** Setup R1

Privilege groups users inherit.

Actors	Lab administrator
Key data	Name (EN/AR), description, status; per screen Query/Insert/Update/Delete; special rights (discount with cap, payment, refund, transfer, cancel, modify verified result, preview without verify, print invoice, VIP…); result rights See/Review/Verify/Edit/Print per clinical group and branch; locations.
Actions	Add, edit, grant/revoke all, filter screens.
Business rules	Active role needs a screen; role with users cannot be deleted; changing a role changes all its users (BR-17).
Depends on	Clinical groups, Branches

CP-38 **Users** Setup R1

Staff accounts.

Actors	Lab administrator
Key data	Name (EN/AR), username, mobile, email, designation, role, branches, shifts, status; password policy, expiry, must-change, lock; signature image; privilege overrides with restore.
Actions	Add, edit, lock, restore to role.
Business rules	Username 4–30 chars; overrides affect one user only; main administrator protected.
Depends on	Roles, Branches

CP-39 **Clinical groups** Setup R1

Laboratory units that split tubes, rights and reports.

Actors	Lab administrator
Key data	Code, unit code, EN/AR names, report name, report template, display order, status.
Actions	Add, edit, deactivate.
Business rules	Group with services cannot be deleted.
Depends on	Report templates

CP-40 **Services** Setup R1

Orderable items: tests/profiles, packages, cultures, text reports, additional services.

Actors	Lab administrator
Key data	Code, short name, shortcut no., EN/AR names, report name, LOINC, CPT, group, type, sample type and condition, header, aliases, precaution, gender, duration, labels, print-alone, visibility flags, status; members with order, in-sum and visibility; profile sum; base price.
Actions	Add, edit, manage members.
Business rules	At least one member; unique member order; package members are services; print-alone services print on their own sheet.
Depends on	Tests, Sample types, Clinical groups, Base price lists

CP-41 Tests Setup R1

Measured parameters and their configuration.

Actors	Lab administrator, unit head
Key data	Code, EN/AR names, report name, LOINC, CPT, group, type (numeric, non-numeric, optional list, restricted list, formula, formatted text, culture), sample type and condition, unit, gender, methodology, decimals, flags; parent services (the services that carry the test, with its order in each); numeric range bands by sex/age/condition with normal and panic limits, reportable limits; list values with normal/critical/default/colour; formula; text paragraphs; duration, minimum duration, frequency check, delta %, test days; labels, prefix, print reference.
Actions	Add, edit, configure per type, check formula; add the test to a profile or culture, or remove it, from the parent-services block.
Business rules	Default list value must not read as a result (BR-19); formulas validated against test codes; hidden helper tests never print (BR-20); parent services follow the member rules of each service type: a test service or text report keeps its single test and a package takes none (set up from the service), profiles and cultures are joined or left from the test.
Depends on	Clinical groups, Sample types, Units, Patient conditions, Services

CP-42 Service sequence Setup R1

Global display and report order of services.

Actors	Lab administrator
Key data	Sequence per service.
Actions	Edit inline.
Business rules	Unique and greater than zero.
Depends on	Services

CP-43 Sample types Setup R1

Sample types and their conditions.

Actors	Lab administrator
Key data	Code, short name, EN/AR names, conditions (fasting, random, refrigerated…), status.
Actions	Add, edit.
Business rules	Type referenced by tests cannot be deleted.
Depends on	—

CP-44 **Units of measurement** Setup R1

Result units.

Actors	Lab administrator
Key data	Name, status.
Actions	Add, edit.
Business rules	Unit referenced by tests cannot be deleted.
Depends on	—

CP-45 **Patient conditions** Setup R2 seeded in Release 1

Conditions that switch reference ranges.

Actors	Lab administrator
Key data	EN/AR names, status.
Actions	Add, edit.
Business rules	Used by registration and range bands (BR-08).
Depends on	Tests

CP-46 **Preparations** Setup R1

Questions asked at registration.

Actors	Lab administrator
Key data	Code, EN/AR question, type (text, yes/no, option list), options, gender, linked services, status.
Actions	Add, edit.
Business rules	Option list needs at least two options; empty service link = all services.
Depends on	Services

CP-47 **Comment library** Setup R1

Reusable report comments.

Actors	Unit head
Key data	Short name, EN/AR text, group scope, status.
Actions	Add, edit.
Business rules	Inserted into report comments with % search.
Depends on	Clinical groups

CP-48 **Doctors** Setup R1

Referring doctors.

Actors	Lab administrator
Key data	Code, EN/AR name, specialty, occupation, branches, share %, phone, status.
Actions	Add, edit.
Business rules	Share % feeds the doctor-income report.
Depends on	Branches

CP-49 **Diagnoses** Setup R2 seeded in Release 1

ICD codes with linked services.

Actors	Lab administrator
Key data	ICD code, EN/AR name, services, status.
Actions	Add, edit.
Business rules	Choosing a diagnosis at registration adds its services.
Depends on	Services

CP-50 **Sending labs** Setup R2

Contracted laboratories sending samples.

Actors	Lab administrator
Key data	EN/AR name, payer, tier, courier (EN/AR), phone, doctor, status.
Actions	Add, edit.
Business rules	—
Depends on	Payers

CP-51 **Shortcut sets** Setup R2 seeded in Release 1

Typing shortcuts dropping several services.

Actors	Lab administrator
Key data	Code, EN/AR name, services, status.
Actions	Add, edit.
Business rules	Each service keeps its own price; lines remain editable.
Depends on	Services

CP-52 **Test rules** Setup R2 seeded in Release 1

Auto-add rules.

Actors	Lab administrator
Key data	EN/AR name, group, trigger services, added services, status.
Actions	Add, edit, validate.
Business rules	A service cannot be both trigger and added; added services are charged at their own price.
Depends on	Services

CP-53 **Reasons** Setup

R2 seeded in Release 1

Reason lists.

Actors	Lab administrator
Key data	Kind (rejection, cancellation, discount, refund), EN/AR name, status.
Actions	Add, edit.
Business rules	—
Depends on	—

CP-54 **Report templates** Setup

R2 seeded in Release 1

Print templates.

Actors	Lab administrator
Key data	Name, type (results, culture, text, claim, receipt, label, job order), logo, signature, QR, language, header text, status.
Actions	Add, edit.
Business rules	—
Depends on	—

CP-55 **Additional services** Setup

R2 seeded in Release 1

Chargeable non-test services.

Actors	Lab administrator
Key data	Code, short name, EN/AR name, shortcut no., status, base price.
Actions	Add, edit.
Business rules	No sample and no result; not on the job order.
Depends on	Base price lists

CP-56 **Drug types · Antibiotics · Organisms** Setup R1

Microbiology masters for culture entry.

Actors	Microbiologist, lab administrator
Key data	Drug types; antibiotics (code, name, abbreviation, commercial name); organisms (code, name).
Actions	Add, edit.
Business rules	—
Depends on	—

CP-57 **Parameters** Setup R2 Release 1 runs on the defaults

Tenant behaviour switches.

Actors	Lab administrator
Key data	Mandatory mobile / doctor / national ID; similar-name warning; discount cap; accession format; auto-print labels / job order / receipt; print unverified; label copies; result language; auto-collect; auto-receive; new barcode on reject; review before verify; delay thresholds; session timeout.
Actions	Toggle; edit.
Business rules	Read by every module at runtime.
Depends on	—

4.8 Integration (5)

CP-58 **Analysers** Integration R2

Connected instruments.

Actors	Integration engineer
Key data	Code, name, group, default sample, mode (uni, bi-directional, download, shared folder), serial or TCP settings, auto-print, download orders, append unregistered tests/samples, overwrite reviewed/verified, print copies, flags (stop / record), status, connection.
Actions	Add, edit, test connection, open mappings.
Business rules	Serial settings must look like 9600,n,8,1; flags marked "stop" hold results in auto-verification.
Depends on	Clinical groups, Sample types, Site agent

CP-59 **Analyser tests** Integration R2

Maps instrument codes to tests.

Actors	Integration engineer
Key data	Per mapping: test, analyser code, host code, result selection, slope, intercept, unit, query, query details, Auto Send, conversion table, analyser-specific range, methodology.
Actions	Map test; edit inline; conversion table; range override; print code sheet.
Business rules	Auto Send is the permission to save incoming results; a mapping without it never lands a result.
Depends on	Analysers, Tests

CP-60 **Analyser samples** Integration R2

Sample-type codes per instrument.

Actors	Integration engineer
Key data	Analyser, sample type, code.
Actions	Add, edit.
Business rules	—
Depends on	Analysers, Sample types

CP-61 **Site agent** Integration R2

Per-branch connector status and logs.

Actors	Integration engineer, support
Key data	Agent name, branch, version, status, heartbeat, messages and errors, connected analysers, configuration, decrypted hourly log.
Actions	Push configuration; fetch logs.
Business rules	Happy path shows StartSend then Passed Check per sample ID; an Exception line names the cause.
Depends on	Analysers, Branches

CP-62 **PCR kits** Integration R2

Interpretation per instrument and kit.

Actors	Integration engineer
Key data	Kit, instrument, serial lock, run-file layout, value column, per-gene positive/borderline thresholds, verdict rules, result/backup/failed folders, status.
Actions	Add, edit, view, simulate run.
Business rules	A low Ct is positive; all genes negative including the internal control = INVALID; files from other instruments are refused.
Depends on	Analysers

4.9 Automation (2)

CP-63 **Verification settings** Automation R2

Auto-verification engine switches.

Actors	Unit head, lab administrator
Key data	Master switch; per clinical group the checkers (medical rules, no-rules guard, result, comment, option, flag, device flag, incomplete, panic, normal, delta, all-group-complete, inter-analyser delta, reportable, QC); result options; result dictionary.
Actions	Enable/disable; toggle checkers; simulate rules.
Business rules	Normal-range checker overrides rules and stays off by default; enable only after simulation, group by group.
Depends on	Medical rules, Tests, Analysers

CP-64 **Medical rules** Automation R2

Release rules per test.

Actors	Unit head
Key data	Group, test, parameters; rules with formula condition, formula, recommendation, active; operator keypad.
Actions	New rule; edit; check rule; print rules.
Business rules	Percentages written as multiplications of the live range (BR-21); every boundary needs its complement (BR-22); last-result rules need No LR (BR-23); a parameter must carry a LOINC code to be usable.
Depends on	Tests, Verification settings

4.10 Account (4)

CP-65 **Profile** Account R1

Laboratory and credential details.

Actors	Lab administrator
Key data	Lab name, portal URL, country, plan, manager, mobile, email; username, password.
Actions	Save; update credentials via OTP.
Business rules	—
Depends on	Subscriptions (SA)

CP-66 **Wallet** Account R2

Prepaid balance and usage.

Actors	Lab administrator
Key data	Balance, next charge, SMS used, transactions.
Actions	Request top-up.
Business rules	Metered services pause at zero; the portal keeps working.
Depends on	Subscriptions (SA)

CP-67 **Support** Account R2

Support tickets.

Actors	Lab administrator
Key data	Ticket no., subject, priority, status, created.
Actions	New ticket.
Business rules	—
Depends on	—

CP-68 **Audit log** Account R2

Tenant audit trail.

Actors	Lab administrator
Key data	Time, user, action, entity, details, branch.
Actions	Filter; export.
Business rules	Immutable.
Depends on	All pages

5. Super Admin Portal — pages

5.1 Access (1)

SA-01 **Login** Access R1

Authenticates National Technology staff.

Actors	Super Admin staff
Key data	Username, password.
Actions	Login.
Business rules	Lock and expiry respected.
Depends on	Admin users

5.2 Overview (2)

SA-02 **Home** Overview R1

Landing page with the brand.

Actors	Super Admin staff
Key data	—
Actions	—
Business rules	—
Depends on	—

SA-03 **Dashboard** Overview R1

Platform KPIs.

Actors	Super Admin staff
Key data	Clients active/inactive, admin users, served patients, tests and registrations per period, wallets at zero, subscription vs manual, plan distribution, registrations by client, recent clients.
Actions	Change period; refresh.
Business rules	—
Depends on	Clients Subscriptions

5.3 Administration (6)

SA-04 **Clients** Administration R1

Tenant lifecycle.

Actors	Operations Admin
Key data	Name (EN/AR), country, governorate, address, mobile, licence, manager name and mobile, email, plan, portal URL, wallet, branches and users counts, created, source (subscription / manual), status.
Actions	Add; edit; reset password; open wallet; activate / deactivate; filters by source and plan.
Business rules	Creating a client copies the global master data (BR-25); deactivation blocks all client users at login.
Depends on	Countries, Governorates, Plans, Master data

SA-05 **Subscriptions & wallets** Administration R1

Plans, wallet balances and transactions.

Actors	Operations Admin
Key data	Plans (fee, branches, users, SMS quota, analysers); client wallet balance, next charge, transactions (top-up, subscription, usage).
Actions	Top up; select client.
Business rules	Monthly fee and metered usage debit the wallet; zero balance pauses metered services.
Depends on	Clients

SA-06 **Roles** Administration R1

Platform roles.

Actors	Super Admin
Key data	Name, description, status; per-screen Query/Insert/Update/Delete; platform special rights (impersonate client admin, reset client password, activate client, expire passwords).
Actions	Add, edit.
Business rules	Active role needs a screen; role with users protected; no per-user override.
Depends on	—

SA-07 **Admin users** Administration R1

National Technology staff accounts.

Actors	Super Admin
Key data	Full name, username, mobile, email, role, password expiry, lock, status.
Actions	Add, edit, lock.
Business rules	Username 4–30 chars; password policy; privileges inherited from the role only; main administrator protected.
Depends on	Roles

SA-08 **Sessions** Administration R1

Active admin sessions.

Actors	Super Admin
Key data	User, IP, since, status.
Actions	Terminate.
Business rules	—
Depends on	Admin users

SA-09 **Audit log** Administration R1

Platform audit trail.

Actors	Super Admin
Key data	Time, user, action, entity, details, client.
Actions	Filter; export.
Business rules	Immutable.
Depends on	All pages

5.4 Master data (4)

SA-10 **Countries** Master data R1

Countries and their validation data.

Actors	Operations Admin
Key data	Name (EN/AR), code, dial code, currency and code, language, time zone, mobile length, flag, remarks, status.
Actions	Add, edit.
Business rules	Dial code like +20; country referenced by governorates cannot be deleted.
Depends on	—

SA-11 **Governorates** Master data R1

Governorates and districts.

Actors	Operations Admin
Key data	Name (EN/AR), country, district, remarks, status.
Actions	Add, edit.
Business rules	—
Depends on	Countries

SA-12 **Clinical groups · Services · Tests · Service sequence · Sample types · Units · Preparations · Comment library**

Master data R1

Global catalogue templates with the same editors as the client portal.

Actors	Master Data Reviewer, Operations Admin
Key data	As CP-39 to CP-47.
Actions	Add, edit, publish.
Business rules	Changes reach tenants according to the master-data update policy (SA-33).
Depends on	Platform parameters

SA-13 **Patient conditions · Reasons · Diagnoses · Additional services · Report templates · Test rules · Shortcut sets**

Master data R1

Global versions of the catalogue-depth lists.

Actors	Operations Admin
Key data	As CP-45, CP-49, CP-51 to CP-55.
Actions	Add, edit, publish.
Business rules	Same publication policy.
Depends on	Services Tests

5.5 Microbiology (1)

SA-14 **Drug types · Commercial names · Antibiotics · Organisms · Sensitivity values** Microbiology R1

Global microbiology masters.

Actors	Master Data Reviewer
Key data	Drug types; commercial names with drug type; antibiotics (code, name, abbreviation, commercial name); organisms (code, name); sensitivity values (S/I/R).
Actions	Add, edit.
Business rules	—
Depends on	—

5.6 Integration library (4)

SA-15 **Analyser models** Integration library R2

Vendor/model library tenants choose from.

Actors	Integration team
Key data	Vendor, model, discipline, protocol, physical link, modes, driver version, status.
Actions	Add, edit.
Business rules	—
Depends on	—

SA-16 **PCR kits** Integration library R2

Kit interpretation library.

Actors	Integration team
Key data	As CP-62.
Actions	Add, edit, view.
Business rules	—
Depends on	Analyser models

SA-17 **Code systems** Integration library R2

LOINC, CPT, ICD-10, SNOMED versions.

Actors	Integration team
Key data	Name, version, items, use, status.
Actions	Import codes.
Business rules	—
Depends on	—

SA-18 **Rule templates** Integration library R2

Reusable medical-rule patterns.

Actors	Integration team
Key data	Name, group, formula, description, status.
Actions	Add, edit.
Business rules	—
Depends on	—

5.7 Platform (3)

SA-19 **Platform parameters** Platform R1

Platform-wide defaults.

Actors	Super Admin
Key data	OTP length and validity, password minimum, session timeout, trial days, default currency and language, master-data update policy, default delay thresholds and discount cap.
Actions	Save.
Business rules	The master-data policy decides whether later global changes are pushed, proposed or ignored for tenants.
Depends on	—

SA-20 **Notification templates** Platform R1

SMS and email texts.

Actors	Super Admin
Key data	Code, channel, EN/AR text with placeholders, status.
Actions	Add, edit.
Business rules	Placeholders: {code} {url} {patient} {acc} {lab} {test} {value}.
Depends on	—

SA-21 **Profile** Platform R1

Administrator details and credentials.

Actors	Super Admin staff
Key data	Name, username, mobile, email; password.
Actions	Save; update credentials via OTP.
Business rules	—
Depends on	Admin users

6. Status models

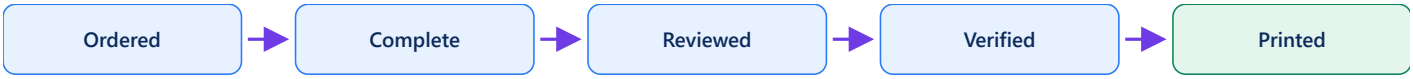
Sample (tube)



Rejected (with reason) → Unrejected or Replaced with a new barcode

Collected on scan at phlebotomy, Received on scan at the unit; auto-collect / auto-receive parameters set both on save. Results need Received (BR-03).

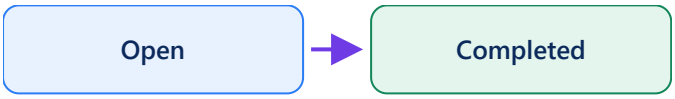
Test / result



Undo returns Verified/Reviewed to the previous state (audited)

A report unit takes the lowest status of its results; a profile advances only when every member has (BR-04). Only Verified prints (BR-09).

Visit



Services may be Reserved (deferred) or Cancelled (with reason)

Completed when every report unit is printed and the balance is settled.

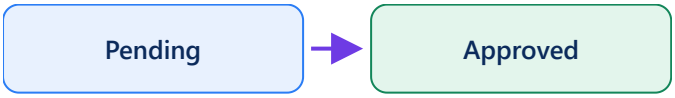
Payment balance



Negative balance → Refund due

Remaining = patient share − payments + refunds (BR-14).

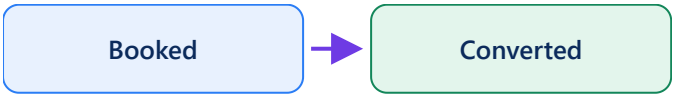
Treasury request



Rejected · Revoked

Bound to a device ID; approval by a manager.

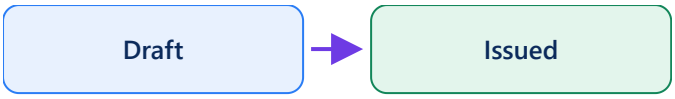
Home booking



Cancelled

Conversion creates the visit.

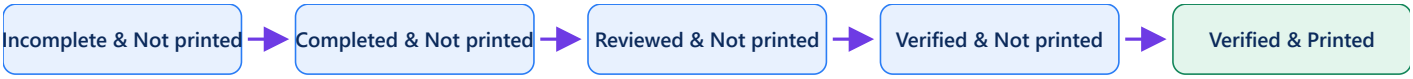
Claim



—

Issued claims are printed and sent to the payer.

Work-queue colour states



Incomplete & Printed · Completed & Printed · Reviewed & Printed · Reserved & Not printed

Nine colours per installation; the meaning is fixed.

7. End-to-end workflows

7.1 WF-01 — Laboratory onboarding

Pages: CP-02 SA-04 CP-36 CP-37 CP-38 CP-33 CP-32 Release: R1

#	Actor	Action	System response
1	**Laboratory manager**	Completes the subscription wizard (lab details, OTP, credentials).	Creates the tenant, copies the global master data, generates the portal URL and sends it by SMS.
2	**Operations Admin (alternative)**	Creates the client manually and activates it.	Same provisioning; source marked Manual.
3	**Lab administrator**	Logs in, selects the branch, creates branches, roles and users.	Users inherit role privileges; treasury requests are raised per device.
4	**Lab administrator / accounts**	Reviews the inherited catalogue, sets base prices and payers with tiers.	Contract prices derive by formula on Apply.
5	**Reception**	Registers the first patient.	Labels, job order and receipt print; the laboratory is live.

7.2 WF-02 — Walk-in patient — registration to receipt

Pages: CP-06 CP-10 CP-23 Release: R1

#	Actor	Action	System response
1	**Reception clerk**	Types the patient name.	Shows similar existing patients with number and mobile (BR-01).
2	**Reception clerk**	Chooses the existing file or confirms a new patient; completes title, gender, date of birth or age, mobile, condition.	Title fills gender; age and birth date sync; patient number assigned on save.
3	**Reception clerk**	Sets payer and contract tier, doctor, flags, diagnosis.	Diagnosis adds its service set; prices follow the contract (BR-06).
4	**Reception clerk**	Adds services by name, number, set or picker.	Refuses duplicates and gender-restricted services; alerts or blocks repeats (BR-12); shows precautions and contract alerts; adds companion services by rule; shows patient and payer shares and expected result dates.

#	Actor	Action	System response
5	**Reception clerk**	Sets a service to Reserved if the precaution is unmet; applies a discount with reason; answers preparation questions.	Reserved lines stay billed without a sample (BR-05); discount limited by the user cap.
6	**Reception clerk (R1)**	Chooses the payment method and amount.	Defaults to the amount due; partial amounts allowed; reference required for non-cash.
7	**Reception clerk**	Saves (Ctrl+S).	Creates the accession, generates tubes by group + sample type + condition (BR-02), creates result rows, records the payment with treasury and device, prints labels, job order and receipt per parameters; shows the success dialog.
8	**Reception clerk (R2)**	Collects the payment from the success dialog or the Payments page.	Receipt printed; balance updated.

7.3 WF-03 — Home visit

Pages: CP-07 CP-06 Release: R2

#	Actor	Action	System response
1	**Reception clerk**	Books the visit with address, time, phlebotomist and services.	Booking number; preliminary labels and job order; quote.
2	**Phlebotomist**	Visits the patient and collects the tubes.	—
3	**Reception clerk**	Converts the booking.	Registration pre-filled; collector set to the phlebotomist; booking marked Converted.

7.4 WF-04 — Batch and lab-to-lab registration

Pages: CP-08 CP-09 CP-27 CP-31 Release: R2

#	Actor	Action	System response
1	**Reception clerk**	Enters the group contract and member rows, or chooses the sending lab and its rows.	One accession per row under a batch number; collected-outside flag on lab-to-lab rows.
2	**Reception clerk**	Saves the batch.	Labels and job orders for all members.
3	**Cashier / accounts**	Settles the batch or lets claims bill the sending lab.	Pay-all distributes across members; lab batches appear on claims.

7.5 WF-05 — Sample lifecycle

Pages: CP-14 CP-15 CP-16 CP-17 Release: R2

#	Actor	Action	System response
1	**Phlebotomist**	Scans the tube after drawing.	Status Collected with time, user and location.
2	**Technician**	Scans the tube at the unit.	Status Received; results may now be entered (BR-03).
3	**Technician**	Rejects an unfit tube with a reason.	Status Rejected; reception notified to recollect; tests take no results.
4	**Technician**	Reinstates or replaces the tube.	Same barcode back to Received, or a new sample with a new barcode (BR-18).
5	**Anyone**	Opens sample tracking.	Timeline, time between stages, test round times.

7.6 WF-06 — Result entry, review, verification and printing

Pages: CP-18 CP-19 CP-20 CP-21 Release: R1 (R2 for sample view and queue)

#	Actor	Action	System response
1	**Technician**	Opens the worklist for the unit.	One row per test with expected result date and delay colour (BR-13).
2	**Technician**	Opens the visit on Results by accession (or scans the tube).	Filters collapse; one tab per report unit with its sample IDs and status.
3	**Technician**	Enters values, picks list values, leaves formulas to compute.	Ranges resolved by sex, age and condition (BR-08); flags and panic alerts; Restricted lists validated; delta flag against the previous result; status Complete on save.
4	**Technician**	Adds a report comment from the library; writes an internal note if needed.	Report comment prints; notes never print.
5	**Technician / unit head**	Reviews, then verifies.	Rights per clinical group checked; verify refused without a value; delta flag must be acknowledged (BR-11); electronic signature recorded.
6	**Reception / technician**	Previews and prints.	Preview refused while values are missing; only verified prints (BR-09); printed flag and status log updated; print list number assigned in bulk printing.

7.7 WF-07 — Analyser and PCR results

Pages: CP-58 CP-59 CP-61 CP-62 CP-21 Release: R2

#	Actor	Action	System response
1	**Integration engineer**	Defines the analyser, maps its codes with Auto Send, configures the site agent.	Agent connects over serial, TCP or file share.
2	**Analyser**	Sends results for a scanned sample ID.	Results land as Complete with the analyser source; flags recorded; conversion tables applied; unmapped or Auto-Send-off codes refused and logged.
3	**PCR instrument**	Exports a run file to the result folder.	Kit interpretation produces DETECTED / NOT DETECTED / Presumptive / INVALID per sample; file moved to backup or failed.
4	**Chemist**	Closes the analyser's work from the queue (from-analyser, then to-analyser).	Everything produced is reviewed; everything pending is traced.

7.8 WF-08 — Auto-verification rollout

Pages: CP-63 CP-64 Release: **R2**

#	Actor	Action	System response
1	**Unit head**	Gives the release rules per group in medical language.	—
2	**Lab administrator**	Writes the rules with the keypad, using formula conditions and No LR where needed; checks each rule.	Rule validated (BR-21 to BR-23).
3	**Lab administrator**	Simulates the rules on recent results.	Released / held counts with reasons.
4	**Lab administrator**	Enables the engine for one group at a time.	Incoming results pass the checkers; normal results release; the rest reach the doctor.

7.9 WF-09 — Money movements and daily closing

Pages: CP-23 CP-24 CP-25 CP-26 CP-29 CP-30 Release: **R1 (payments, treasury) · R2 (rest)**

#	Actor	Action	System response
1	**Cashier**	Requests treasury access from the device; manager approves.	Payments possible on that treasury/device.
2	**Cashier**	Collects full or partial payments.	Receipt per payment; remaining balance tracked.
3	**Accounts**	Transfers a visit to a contract after approval arrives, or shifts a line value.	Re-pricing; excess becomes refundable.

#	Actor	Action	System response
4	**Cashier**	Refunds a negative balance with reason.	Refund receipt for patient signature; audited.
5	**Reception**	Cancels a service with reason.	Receivables adjusted; samples cancelled; record kept (BR-15).
6	**Cashier**	Closes the drawer.	Expected cash = cash payments − refunds (BR-16); handover sheet printed.

7.10 WF-10 — Pricing maintenance and claims

Pages: CP-33 CP-32 CP-34 CP-31 CP-35 Release: **R1 (lists, payers) · R2 (pricing page, claims)**

#	Actor	Action	System response
1	**Accounts**	Creates next year's base list by copying with a factor.	All prices derived in one step.
2	**Accounts**	Points each tier at the new list with patient and payer factors and applies.	Contract prices refresh; excluded prices kept (BR-07).
3	**Accounts**	Issues the monthly claim per credit payer.	Visits and payer shares listed in the chosen layout; draft then issued.
4	**Management**	Runs the financial reports.	Income, debtors, discounts, cancellations, turnaround.

7.11 WF-11 — Privileges administration

Pages: CP-37 CP-38 Release: **R1**

#	Actor	Action	System response
1	**Lab administrator**	Agrees the team split with the lab manager.	—
2	**Lab administrator**	Creates roles with screen rights, special rights and result rights per group and branch.	Users joined to a role inherit everything (BR-17).
3	**Lab administrator**	Adjusts one user if needed.	Marked modified; restore returns the user to the role.

7.12 WF-12 — Platform operations

Pages: SA-04 SA-05 SA-12 SA-13 SA-19 Release: **R1**

#	Actor	Action	System response
1	**Operations Admin**	Creates or activates clients, resets passwords, tops up wallets.	Audit entries; client users blocked when inactive.

#	Actor	Action	System response
2	**Master Data Reviewer**	Maintains the global catalogue and lists.	Changes published to tenants according to the policy.
3	**Super Admin**	Sets platform parameters and notification templates.	OTP, password and session rules applied platform-wide.

8. Business rules

ID	Rule	Statement
BR-01	Immutable identifiers	Patient numbers are generated by the system and never change; accessions are sequential per visit; sample IDs are unique and never reused.
BR-02	Tube merge	Services with the same clinical group, sample type and sample condition share one sample; any difference splits them.
BR-03	Received gate	A result may not be recorded until the sample status is Received (auto-receive emulates it in Release 1).
BR-04	Profile completion	A profile or package reaches Complete, Reviewed or Verified only when every member test has.
BR-05	Reserved keeps the bill	Deferring a service keeps it in the visit and the bill with no sample; cancelling removes it from both.
BR-06	Contract sets the price	Payer and contract tier determine every service's patient share and payer share.
BR-07	Three-layer pricing	Base list price × tier factors = contract price; Apply propagates base-list changes to all derived tiers except excluded services.
BR-08	Patient condition selects the range	Reference ranges resolve by sex, age and patient condition.
BR-09	Verify is a signature	Verification records who signed and when; only verified results print.
BR-10	Not-X queue logic	"Not X" returns everything below X in the test track.
BR-11	Delta check warns	A delta flag must be investigated and acknowledged but does not block verification.
BR-12	Frequency check	A repeat order inside the configured interval alerts or is prevented.
BR-13	Turnaround clock	Expected result = collection + duration counting only test days; delay colours are a percentage of the duration consumed.
BR-14	Negative balance means refund	A negative visit balance is returned through the refund flow, never above the refundable amount.
BR-15	Nothing is erased	Cancellations, transfers, rejections and edits keep the original data with reason, user and time.

ID	Rule	Statement
BR-16	Drawer equation	Expected cash = cash payments − refunds; card and transfer amounts are not in the drawer.
BR-17	Group privileges propagate	Changing a role changes every user in it; changing one user affects that user only.
BR-18	New sample, new barcode	A replacement sample carries a new barcode unless the tenant policy allows reinstatement.
BR-19	Default list value is not a result	The default of a list test is empty or a placeholder, never a value that reads as a result.
BR-20	Hidden helpers	Helper tests feeding formulas or sums never print.
BR-21	Percentage tolerance is a multiplication	A ±20 % tolerance is written as × 0.8 and × 1.2 of the live range values.
BR-22	Every boundary needs its complement	A rule for one side of a boundary needs a condition or complement so other patients are not stopped.
BR-23	No last result must not stop	Rules using the previous result must not stop patients who have none in the window.
BR-24	Validated selection only	Fields that reference master data accept only selected values.
BR-25	Tenant copy of master data	A tenant's master data is an independent copy of the global templates; later global changes follow the update policy.
BR-26	Payment inside registration (Release 1)	The amount captured on the registration page is recorded with the visit on save; it may not exceed the amount due; non-cash payments need a reference.
BR-27	Treasury per device	A cash transaction is accepted only on a treasury approved for the user's device.

9. Conceptual data model

Area	Entities
Platform	Client (tenant) · Plan · Wallet transaction · Admin user · Admin role · Country · Governorate · Notification template · Platform parameter
Tenant foundation	Branch · User · Role / privilege · Treasury · Treasury request (device) · Parameter · Audit entry
Catalogue	Clinical group · Sample type · Sample condition · Unit · Service · Service member · Test · Reference range band · List value · Formula · Preparation question · Comment template · Report template · Doctor · Diagnosis · Shortcut set · Test rule · Reason · Patient condition · Sending lab · Organism · Antibiotic · Sensitivity value
Operations	Patient · Patient document · Visit (accession) · Order line · Preparation answer · Sample · Sample event · Result · Result log entry · Status log entry · Culture result · Text result · Attachment · Print list · Booking · Batch
Finance	Payer · Contract tier · Base price list · Contract pricing · Price line · Payment · Refund · Transfer · Cancellation · Cash closing · Claim
Integration & automation	Analyser · Analyser test mapping · Analyser sample code · Site agent · Agent log · PCR kit · Verification settings · Medical rule · Result option · Result dictionary word

Relationships: a tenant has branches, users, roles and a master-data copy · a patient has visits · a visit has order lines, samples, payments, refunds and documents · a sample has status events and labels · an order line yields results with logs, flags, comments and attachments · a payer has contract tiers whose prices derive from base lists · an analyser has mappings and agent logs · rules belong to tests.

10. Documents and reports

Document	Content	Produced from
Barcode label	Per tube: unit code, date/time, location, sample type and condition, accession, patient, sample ID barcode, services.	Registration save, Samples by visit
Job order	Visit header, balance due, per service unit / expected result / sample / barcode, collection boxes; "collected outside" note.	Registration save, Samples by visit
Receipt / invoice	Payment receipt per payment; visit invoice.	Registration (R1), Payments
Refund receipt	Amount, method, reason, signature boxes.	Refunds
Price quote	Services with shares, expected dates, preparations.	Price query
Laboratory report	Template, patient and visit header, results with units, ranges and flags per report unit, comments, verifier signature, QR.	Results, Work queue, Print reports
Abbreviated worksheet	Multi-patient file copy with empty result boxes.	Work queue
Worklist	Bench list of tests with expected dates.	Worklist
Cash-closing sheet	Payments, refunds, expected cash, handover signatures.	Cash closing
Claim	Payer visits and shares in the chosen layout with totals and approval boxes.	Claims
Analyser code sheet	Mapped codes per analyser.	Analyser tests
Financial reports	Sixteen reports with period and branch filters, print and Excel.	Financial reports

11. Non-functional requirements (summary)

Quality	Requirement
Bilingual & RTL	Arabic and English on every screen and document with true right-to-left layout.
Responsive	Desktop, tablet and phone; the navigation collapses to a drawer on phones.
Security	Tenant isolation, TLS, encrypted health data at rest, OTP channels, password policy, role-based access, device-bound cash, session timeout, audit trail.
Performance	1,500 samples per shift per unit; results screen under 2 s; searches under 3 s; labels within 2 s of save.
Availability	99.9 % monthly target with daily backups and point-in-time recovery.
Auditability	Immutable audit records; verification is an electronic signature.
Printing	Built-in Code 128 and QR rendering, PDF generation, printer selection per workstation.
Configurability	Tenant parameters change behaviour without code changes.
Data-entry speed	Keyboard-operable high-volume screens with shortcuts.
Migration	Path from LDM for patients, visits, results, catalogue, prices, contracts and users.

The full non-functional set is in LIS-RS-001, section 4.

12. Release scope matrix

12.1 Client Portal

ID	Page	Module	Release 1	Note
CP-01	**Public website**	Access & onboarding	Yes	
CP-02	**Subscription wizard**	Access & onboarding	Yes	
CP-03	**Login & password recovery**	Access & onboarding	Yes	
CP-04	**Branch selection**	Access & onboarding	Yes	
CP-05	**Dashboard**	Overview	Yes	
CP-06	**Registration**	Reception	Yes	without Location/Department and Room; payment captured before saving
CP-07	**Home reservation**	Reception	Release 2	
CP-08	**Batch registration**	Reception	Release 2	
CP-09	**Lab-to-Lab registration**	Reception	Release 2	
CP-10	**Search registrations**	Reception	Yes	
CP-11	**Patients**	Reception	Yes	
CP-12	**Price query**	Reception	Yes	
CP-13	**Print reports**	Reception	Release 2	
CP-14	**Collection & receipt**	Samples	Release 2	in Release 1 samples are received automatically on save
CP-15	**Samples by visit**	Samples	Release 2	

ID	Page	Module	Release 1	Note
CP-16	**Reject / unreject**	Samples	Release 2	
CP-17	**Sample tracking**	Samples	Release 2	
CP-18	**Worklist**	Laboratory	Yes	
CP-19	**Results by accession**	Laboratory	Yes	filters auto-collapse, report units as tabs, no analyser column
CP-20	**Results by sample**	Laboratory	Release 2	
CP-21	**Work queue**	Laboratory	Release 2	
CP-22	**Quality control**	Laboratory	Release 2	
CP-23	**Payments**	Finance	Yes	
CP-24	**Refunds**	Finance	Release 2	refund action also available on Payments in R1
CP-25	**Money transfer**	Finance	Release 2	
CP-26	**Cancellations**	Finance	Release 2	available as an action on Search registrations in R1
CP-27	**Batch payment**	Finance	Release 2	
CP-28	**Treasuries**	Finance	Yes	
CP-29	**Treasury requests**	Finance	Yes	
CP-30	**Cash closing**	Finance	Release 2	
CP-31	**Claims**	Finance	Release 2	
CP-32	**Payers & contracts**	Finance	Yes	

ID	Page	Module	Release 1	Note
CP-33	**Base price lists**	Finance	Yes	
CP-34	**Contract pricing**	Finance	Release 2	Release 1 uses the seeded contract formulas
CP-35	**Financial reports**	Finance	Release 2	
CP-36	**Branches**	Setup	Yes	
CP-37	**Roles & privileges**	Setup	Yes	
CP-38	**Users**	Setup	Yes	
CP-39	**Clinical groups**	Setup	Yes	
CP-40	**Services**	Setup	Yes	
CP-41	**Tests**	Setup	Yes	
CP-42	**Service sequence**	Setup	Yes	
CP-43	**Sample types**	Setup	Yes	
CP-44	**Units of measurement**	Setup	Yes	
CP-45	**Patient conditions**	Setup	Release 2	seeded in Release 1
CP-46	**Preparations**	Setup	Yes	
CP-47	**Comment library**	Setup	Yes	
CP-48	**Doctors**	Setup	Yes	
CP-49	**Diagnoses**	Setup	Release 2	seeded in Release 1

ID	Page	Module	Release 1	Note
CP-50	**Sending labs**	Setup	Release 2	
CP-51	**Shortcut sets**	Setup	Release 2	seeded in Release 1
CP-52	**Test rules**	Setup	Release 2	seeded in Release 1
CP-53	**Reasons**	Setup	Release 2	seeded in Release 1
CP-54	**Report templates**	Setup	Release 2	seeded in Release 1
CP-55	**Additional services**	Setup	Release 2	seeded in Release 1
CP-56	**Drug types · Antibiotics · Organisms**	Setup	Yes	
CP-57	**Parameters**	Setup	Release 2	Release 1 runs on the defaults
CP-58	**Analysers**	Integration	Release 2	
CP-59	**Analyser tests**	Integration	Release 2	
CP-60	**Analyser samples**	Integration	Release 2	
CP-61	**Site agent**	Integration	Release 2	
CP-62	**PCR kits**	Integration	Release 2	
CP-63	**Verification settings**	Automation	Release 2	
CP-64	**Medical rules**	Automation	Release 2	
CP-65	**Profile**	Account	Yes	
CP-66	**Wallet**	Account	Release 2	

ID	Page	Module	Release 1	Note
CP-67	**Support**	Account	Release 2	
CP-68	**Audit log**	Account	Release 2	

12.2 Super Admin Portal

ID	Page	Module	Release 1	Note
SA-01	**Login**	Access	Yes	
SA-02	**Home**	Overview	Yes	
SA-03	**Dashboard**	Overview	Yes	
SA-04	**Clients**	Administration	Yes	
SA-05	**Subscriptions & wallets**	Administration	Yes	
SA-06	**Roles**	Administration	Yes	
SA-07	**Admin users**	Administration	Yes	
SA-08	**Sessions**	Administration	Yes	
SA-09	**Audit log**	Administration	Yes	
SA-10	**Countries**	Master data	Yes	
SA-11	**Governorates**	Master data	Yes	
SA-12	**Clinical groups · Services · Tests · Service sequence · Sample types · Units · Preparations · Comment library**	Master data	Yes	
SA-13	**Patient conditions · Reasons · Diagnoses · Additional services · Report templates · Test rules · Shortcut sets**	Master data	Yes	
SA-14	**Drug types · Commercial names · Antibiotics · Organisms · Sensitivity values**	Microbiology	Yes	

ID	Page	Module	Release 1	Note
SA-15	**Analyser models**	Integration library	Release 2	
SA-16	**PCR kits**	Integration library	Release 2	
SA-17	**Code systems**	Integration library	Release 2	
SA-18	**Rule templates**	Integration library	Release 2	
SA-19	**Platform parameters**	Platform	Yes	
SA-20	**Notification templates**	Platform	Yes	
SA-21	**Profile**	Platform	Yes	

13. Open questions

ID	Question
OQ-01	Original Sky LIS SRS referenced by the first prototypes: to be reconciled with this document.
OQ-02	Wallet semantics (prepaid credit, consumption events, behaviour at zero) — modelled here as prepaid credit debited by subscription and metered usage.
OQ-03	Master-data update policy between platform and tenants (push, propose, none).
OQ-04	Whether Release 1 should keep the Received gate (scan-based) instead of auto-receive.
OQ-05	Review/verify order and whether review may be skipped (currently a tenant parameter).
OQ-06	Scope of quality control, outsourcing, SMS and online transfer modules.
OQ-07	Accession and sample ID formats per tenant and label stock sizes.
OQ-08	Analyser protocols and instruments of the pilot laboratory.

13.1 Glossary

Term	Definition
Accession	Visit / order number.
Clinical group	Laboratory unit; splits tubes, rights and report sections.
Contract tier	Sub-category of a payer with its own price list.
Delta check	Comparison with the patient's previous result within a window.
Job order	Printout travelling with the tubes.
Package	Set of services priced as one.
Panic value	Result beyond critical limits requiring immediate notification.
Patient condition	Attribute that selects an alternative reference range.
Payer	Body paying for services: cash, company, syndicate, insurer, sending lab.

Term	Definition
Preparation	Question about the patient's readiness asked at registration.
Received gate	No result before the sample is received.
Report unit	Visit × clinical group (print-alone services separate): what prints as one report.
Reserved	Deferred service kept in the visit without a sample.
Service	Orderable catalogue item.
Treasury	Cash drawer at a branch.
Turnaround (TAT)	Registration → verification (patient) or receipt → verification (lab).
Verification	Doctor's electronic sign-off that releases a result for printing.
Work queue	Status-driven list of report units.